ST. MARY'S DSG, KLOOF

PHYSICAL SCIENCES
Paper 2

GRADE 10

November 2025 **TIME: 2 hours + 10 min reading**
MARKS: 131

Examiner: G. Main
Moderators: K. Bolton and T. Kirsten

NAME: _____ TEACHER'S NAME:_____

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This paper consists of a question paper of **15 pages**. Please check that your question paper is complete. A **data sheet** is also provided.

2. Read the questions carefully.

3. It is in your own interest to write legibly and to present your work neatly.

4. Answer all questions in the spaces provided.

5. **All working must be shown** and units may be used in substitutions, but **MUST** be given with answers. Work to 2 decimal places or significant figures where appropriate.

6. ONE blank page (15) is included at the end of the question paper. If you run out of space for a question, use this extra page. Clearly indicate the question number of your answer, should you use this extra space.

	Q1	Q2	Q3	Q4	Q5	Q6	Total	%
Mark								
Question total	**12**	**21**	**16**	**37**	**20**	**25**	**131**	**100**

QUESTION 1 – MULTIPLE CHOICE

Answer the multiple-choice questions on the answer grid below. Make a clear cross (X) in the box corresponding to the letter that you consider to be correct. Every question has only one correct answer.

A	B	C	D

Here the option C has been marked as an example.

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D

1.1 The correct formula for chromium (III) sulfite is:

A $Cr_3(SO_3)_2$
B $Cr_2(SO_3)_3$
C $Cr_3(SO_4)_2$
D $Cr_2(SO_4)_3$

1.2 A neutral atom of an element has an electron configuration of $1s^2 2s^2 2p^4$.
In which group and period of the periodic table is this element located?

A Group 2, Period 4
B Group 14, Period 2
C Group 16, Period 2
D Group 17, Period 4

1.3 Study the unknown elements J to N below

	J	K	L	M	N
ELEMENT	$^{19}_{9}X$	$^{19}_{10}X$	$^{20}_{9}X$	$^{21}_{11}X$	$^{19}_{8}X$

Which TWO elements from J to N are isotopes?

A J and K
B J and N
C K and M
D J and L

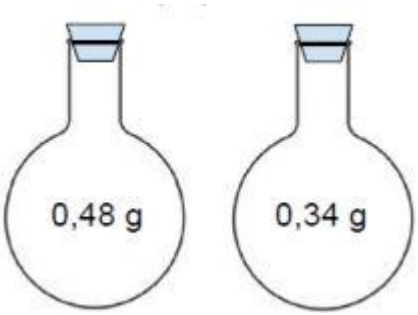
1.4 Consider the unbalanced chemical equation below.

$$A_2(s) + B_3(g) \rightarrow AB_2(g)$$

Which ONE of the sets of coefficients will balance the chemical equation?

A 4 : 6 : 3
B 2 : 3 : 2
C 3 : 4 : 6
D 3 : 6 : 4

1.5 Two identical flasks of identical volume are filled with different gases at the same temperature and pressure. One flask contains ozone (O_3) and the other an unknown gas X.



ozone (O_3) gas X

The mass of the ozone gas in the flask is 0,48 g and the mass of gas X in the flask is 0,34 g.

Gas X is:

A O_2
B SO_2
C H_2S
D NH_3

1.6 Study the redox reaction below.

$$Br_2 + SO_2 + 2H_2O \rightarrow H_2SO_4 + 2HBr$$

Which one of the following combinations is correct?

	Substance oxidised	**Substance reduced**
A	SO_2	H_2O
B	SO_2	Br_2
C	Br_2	SO_2
D	H_2O	Br_2

[6 x 2 = 12]

QUESTION 2

2.1 In the table below are different substances. Use the following substances to answer the questions that follow. You may write the corresponding LETTER down. Letters may be used more than once or not at all.

	SOLID SUBSTANCES
A	Cobalt
B	Gold
C	Phosphorus
D	Silicon dioxide
E	Carbon dioxide
F	Ammonium nitrate

2.1.1 Identify the substance that is an ionic solid. (1)

2.1.2 Identify the substance that is a giant covalently bonded atomic network. (1)

2.1.3 Which element has magnetic properties? (1)

2.1.4 Identify the element that has a dull surface and cannot conduct electricity. (1)

2.1.5 Identify the substance that is a molecular solid. (1)

2.1.6 Write down the chemical formula for substance **F** (Ammonium nitrate). (2)

2.2 Consider the electron configuration of the three elements given below:

ELEMENT	**ELECTRON CONFIGURATION**
P	$1s^2 2s^2 2p^6 3s^2 3p^5$
Q	$1s^2 2s^2 2p^6$
R	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

2.2.1 Write down the NAME of the element represented by electron configuration **R**. (1)

2.2.2 Name of the group to which **Q** belongs. (1)

2.2.3 Which element (**P**, **Q** or **R**) is in group 17, period 3? (1)

2.2.4 Write down the chemical formula of the compound that is formed when **P** reacts with **R**. (2)

2.3 Boron has two stable isotopes. 19% B - 10 and 81% B - 11

2.3.1 Which atomic particle is responsible for the occurrence of isotopes? (1)

2.3.2 Explain how the two isotopes are similar and how they differ by referring to the number and type of particles in each nucleus of the isotope. (2)

2.3.3 Determine the relative atomic mass of the boron sample. (3)

Boron reacts with fluorine to form boron trifluoride.

2.3.4 Draw an Aufbau diagram for a fluoride ION. (3)

[21]

QUESTION 3

The following chemical substances are given: **$MgBr_2$; Fe; CH_4; B, NH_3, $C\ell_2$**

3.1 Identify the type of intramolecular bond formed in the following:

3.1.1 $MgBr_2$ (1)

3.1.2 CH_4 (2)

3.2 Describe the bond you identified in Q3.1.2. (3)

3.3 Which of the chemical substances will form a bond between its positive ions and delocalised valence electrons? (1)

3.4 Give the name of NH_3. (1)

3.5 Draw the Lewis dot diagram for NH_3. (2)

3.6 Name the intermolecular force between NH_3 molecules. (2)

3.7 Draw the Lewis dot diagram to show how $MgBr_2$ is formed. (4)

[16]

QUESTION 4

4.1 Consider the following reaction: **$2H_2O_2 \rightarrow 2H_2O + O_2$**

4.1.1 Circle or highlight the correct term:

The above reaction is a COMBINATION/ DISPLACEMENT/ DECOMPOSITION reaction. (1)

4.1.2 Describe the test for O_2. (2)

4.1.3 Depict a water molecule using Couper notation. (2)

4.1.4 Name and describe the **origin** of the intermolecular forces between oxygen molecules. (3)

4.1.5 Calculate the mass of water that would be produced from 20,4 g of H_2O_2. (5)

4.1.6 Calculate the volume of O_2 produced from the 20,4 g of H_2O_2. (3)

4.2 The UNBALANCED chemical equation below represents chemical change taking place in a container.

$K + CuSO_4 \rightarrow Cu + K_2SO_4$

4.2.1 Balance the above equation. (1)

4.2.2 Give the name of $CuSO_4$. (2)

4.2.3 Circle or highlight the correct term:

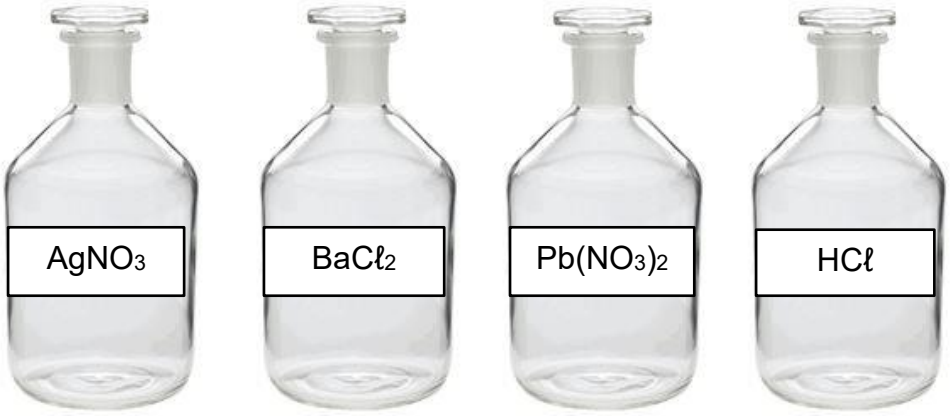
The above reaction is a COMBINATION/ DISPLACEMENT/ DECOMPOSITION reaction. (1)

4.2.4 A sample of K_2SO_4 crystals and a second sample of $CuSO_4$ crystals are placed in a flame. Indicate the colours observed when each sample is placed in the flame. (2)

K_2SO_4	**$CuSO_4$**

4.2.5 Some solid $CuSO_4$ is dissolved in water. Write the balanced dissolution equation for solid $CuSO_4$ dissolved in water. Include all relevant state symbols. (3)

4.3 A learner is provided with four bottles containing different chemical substances as shown in the diagram.



The learner is able to take samples of each substance using a pipette.

4.3.1 Complete the following table showing a successful test for the SULFATE ION. (4)

TEST	CHEMICAL USED	OBSERVATIONS
1		
2		

4.3.2 A LITHIUM SALT solution undergoes the following tests:

TEST	CHEMICAL USED	OBSERVATION
3	Add $AgNO_3$	Pale yellow precipitate
4	Add $Pb(NO_3)_2$	

4.3.2.1 Give the chemical name of $Pb(NO_3)_2$. (2)

4.3.2.2 IN THE TABLE ABOVE, fill in what observation was made in test 4. (1)

4.3.2.3 Write the balanced chemical equation for Test 4, include state symbols. (5)

[37]

QUESTION 5

5.1 An antacid tablet mainly consists of **$NaHCO_3$**. The balanced chemical reaction between an antacid tablet and water is as follows:

$NaHCO_3(s) + H_2O(\ell) \rightarrow NaOH(aq) + H_2O(\ell) + CO_2(g)$

5.1.1 Give the chemical name for **$NaHCO_3$**. (2)

5.1.2 Describe a test for $CO_2(g)$. (3)

5.2 Write down the oxidation numbers of the underlined element in the following particles:

5.2.1 H**N**O_3 (2)

5.2.2 K**Mn**O_4 (2)

5.3 The unbalanced equations **A** and **B** below represent two redox reactions.

A: $FeC\ell_3 + H_2 \rightarrow FeC\ell_2 + HC\ell$

B: $H_2S + Br_2 \rightarrow HBr + S$

5.3.1 Define *oxidation.* (2)

5.3.2 Write down the **FORMULA** of the substance which is:

i) Oxidised in reaction **A**. (2)

ii) The oxidising agent in reaction **B**. (2)

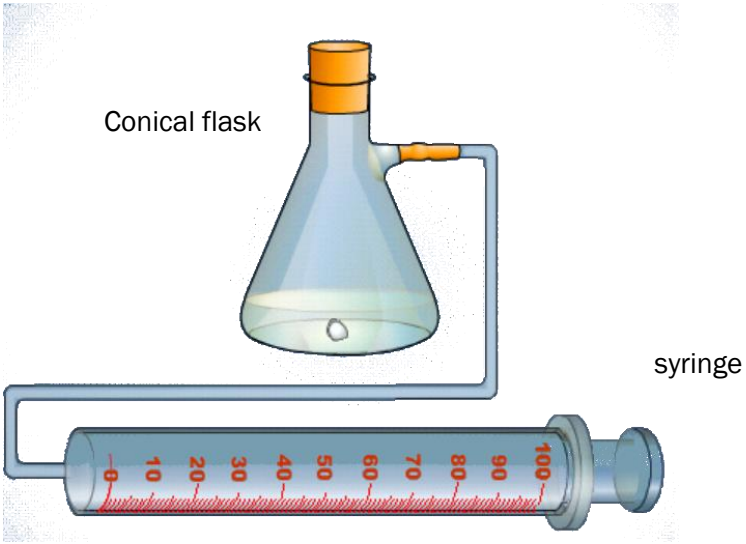
iii) The spectator ion in reaction **B.** (2)

5.3.3 Explain your answer to 5.3.2 i) above in terms of oxidation numbers. (3)

[20]

QUESTION 6

Mbali and Angela conduct an experiment to determine how changing the concentration of an acid affects the volume of carbon dioxide gas produced in a fixed time period with a fixed mass of sodium carbonate. They react hydrochloric acid with the sodium carbonate in a conical flask and then collect the gas in a syringe. They carry out the experiment three times for each concentration of acid.



The reaction that takes place in the conical flask is as follows:

$$Na_2CO_3 + HC\ell \rightarrow NaC\ell + H_2O + CO_2$$

	Volume of gas (dm^3)			
Concentration ($mol.dm^{-3}$)	**Experiment 1**	**Experiment 2**	**Experiment 3**	**Average**
0,1	0,03	0,05	0,07	0,05
0,3	0,15	0,12	0,18	0,15
0,4	0,21	0,19	0,17	0,19
0,6	0,26	**X**	0,29	0,27
0,8	0,44	0,4	0,36	0,4

6.1 Balance the chemical equation for the reaction that takes place in the conical flask. (2)

$$Na_2CO_3 + HC\ell \rightarrow NaC\ell + H_2O + CO_2$$

6.2 Determine the value of **X** and write your answer below. (1)

6.3 Identify the dependent variable in the experiment. (1)

6.4 Identify **TWO** variables which must remain controlled. (2)

6.5 EXPLAIN why the above variables must be controlled. (2)

For ease of reference, the table of results is given again.

	Volume of gas (dm^3)			
Concentration ($mol.dm^{-3}$)	**Experiment 1**	**Experiment 2**	**Experiment 3**	**Average**
0,1	0,03	0,05	0,07	0,05
0,3	0,15	0,12	0,18	0,15
0,4	0,21	0,19	0,17	0,19
0,6	0,26	**X**	0,29	0,27
0,8	0,44	0,4	0,36	0,4

6.6 Use the results in the table to plot a graph of **volume** against **concentration.** (7)



6.7 Using a scientific relationship, state a suitable conclusion for your experiment based on your graph. (3)

6.8 Calculate the gradient of your graph. Make sure you indicate on the graph where you are obtaining your values. (No units are needed in your answer.) (3)

6.9 By referring to the type and strength of intermolecular forces, explain why water is a liquid at room temperature and carbon dioxide is a gas at room temperature. (4)

[25]

EXTRA SPACE